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$$\int \frac{dx}{\sqrt{(1+x^2)^5}}dx \overset{x=\tan u}{=} \int \frac{\sec^2 u}{\left(\tan^2 u + 1\right)^{\frac{5}{2}}}du$$
$$= \int \frac{\sec^2 u}{\sec^5 u}du = \int \frac{1}{\sec^3 u}du = \int \cos u \left(1 - \sin^2 u\right) du$$
$$\overset{v=\sin u}{=} \int \left(1 - v^2\right) dv$$
$$v - \frac{v^3}{3} + c \overset{v=\sin u}{=} \sin u - \frac{\sin^3 u}{3} + c$$
$$\overset{x=\tan u}{=} \frac{x}{\sqrt{x^2+1}} - \frac{x^3}{3\left(x^2+1\right)^{\frac{3}{2}}} + c$$

References